

Problems with modelling commodities

Seasonality: Seasonal peaks driven by production constraints/demand cycles.

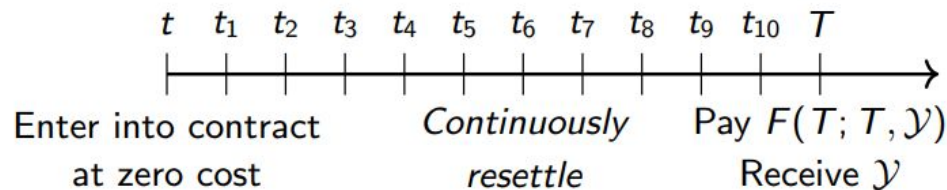
Storage: Storing physical commodities can be expensive.

Convenience: There is an implied 'convenience'/benefit in holding certain commodities

Combine convenience yield and storage cost into net conv. yield δ

High inventory $\rightarrow$ Low convenience Low inventory $\rightarrow$ High convenience

Commodities are usually traded through Futures:



No arbitrage condition:

$$F(t; T, \mathcal{Y}) = \mathbb{E}_t^Q \{ \mathcal{Y} \}.$$

Assuming constant interest rate and net conv. yield, the futures price is:

$$F(t; T) = S(t)e^{(r-\delta)(T-t)}$$

$S(t)$ = Price of commodity. $r - \delta$ = "cost of carry".

Affine Commodity Term structures

Standard Setup:

$$d\mathbf{X}_t = \mu(\mathbf{X}_t)dt + \sigma(\mathbf{X}_t)dW_t^{\mathbb{Q}} \quad r_t = \rho_0 + \rho_1 \cdot \mathbf{X}_t \quad P(t, T) = \mathbb{E}_t^{\mathbb{Q}} \left\{ e^{-\int_t^T (\rho_0 + \rho_1 \cdot \mathbf{X}_u) du} \right\} = e^{A(T-t) + B(T-t) \cdot \mathbf{X}_t}$$

New assumption, commodity prices are exponentially affine to $\mathbf{X}$: $S_t = e^{\delta_0 + \delta_1 \cdot \mathbf{X}_t}$

Futures price is easily determined: $F(t, T) = \mathbb{E}_t^{\mathbb{Q}} \{ S_T \} = \mathbb{E}_t^{\mathbb{Q}} \left\{ e^{\delta_0 + \delta_1 \cdot \mathbf{X}_T} \right\} = e^{A(T-t) + B(T-t) \cdot \mathbf{X}_t}$

Where A and B solve: $\frac{dB(\tau)}{d\tau} = K_1^\top B(\tau) + \frac{1}{2} B^\top(\tau) H_1 B(\tau), B(0) = \delta_1 \quad \frac{dA(\tau)}{d\tau} = K_0 \cdot B(\tau) + \frac{1}{2} B^\top(\tau) H_0 B(\tau) A(0) = \delta_0.$

Schwartz one-factor affine model: $S_t = e^{x_t} \quad dx_t = \kappa(\theta - x_t)dt + \sigma dW^{\mathbb{Q}}(t)$

Very simple solution! $B(\tau) = e^{-\kappa\tau} \quad A(\tau) = (1 - e^{-\kappa\tau})\theta + \frac{\sigma^2}{4\kappa}(1 - e^{-2\kappa\tau})$

Idea: Model δ as OU-process with positive correlation to asset price to capture mean-reversion and connection between convenience and price shocks. This is exactly what Gibson and Schwartz model does.

$$dx_t = \left(r - \delta_t - \frac{\sigma_S^2}{2} \right) dt + \sigma_S dW_1^{\mathbb{Q}}(t) \quad d\delta_t = \kappa(\alpha - \delta_t)dt + \sigma_\delta dW_2^{\mathbb{Q}}(t) \quad \langle W_1^{\mathbb{Q}}, W_2^{\mathbb{Q}} \rangle_t = \rho t$$

Seasonality

One way:

$$S_t = e^{\gamma(t)} e^{\delta_0 + \delta_1 \cdot \mathbf{x}_t}$$

$$\gamma(t) = \sum_{m=1}^M [\gamma_m^c \cos(2\pi mt) + \gamma_m^s \sin(2\pi mt)]$$



Deterministic truncated
Fourier series.

$m = 1 \rightarrow$ yearly frequency, $m = 2 \rightarrow$ half-yearly frequency. Still affine model:

$$F(t, T) = \mathbb{E}_t^{\mathbb{Q}} \{S(T)\} = e^{\gamma(t)} \mathbb{E}_t^{\mathbb{Q}} \left\{ e^{\delta_0 + \delta_1 \cdot \mathbf{x}_T} \right\}$$

Other way:

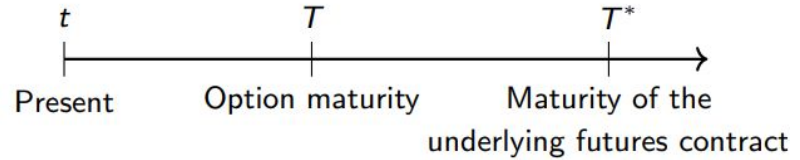
If we want to model different seasonal volatility we could define the volatility for the state variable as:

$$\sigma_{\mathbf{x}}(t) = \sigma_{\mathbf{x}} e^{\gamma(t)}$$

Which does ruin the time-homogeneity of the affine process.

Futures Option:

Payoff: $(F(T, T^*) - K)^+$



Black 76: $\frac{dF_T(t)}{F_T(t)} = \sigma_F dW^{\mathbb{Q}}(t)$ $C = P(t, T) [F(t, T^*)\Phi(d_+) - K\Phi(d_-)]$ $d_{\pm} \triangleq \frac{\log \frac{F(t, T^*)}{K} \pm \frac{\sigma_F^2}{2}(T - t)}{\sigma_F \sqrt{T - t}}$

HJM: $\frac{dF(t, T)}{F(t, T)} = \sigma_F(t, T) \cdot dW_t^{\mathbb{Q}}$ Assume deterministic vol. $F(T, T^*) = F(t, T^*)e^{-\frac{1}{2}\nu^2(t, T, T^*) - \nu(t, T, T^*)Z}$

F is log-normal $\rightarrow$ Use Black76

$$d_{\pm} = \frac{\log \frac{F(t, T^*)}{K} \pm \frac{\nu^2(t, T, T^*)}{2}}{\nu(t, T, T^*)}$$

Stochastic interest rates and convenience yields

$$\begin{array}{lll} \text{ZCB:} & \text{Forwards:} & \text{Futures:} \\ P(t, T) = e^{-\int_t^T f(t,u)du} & F(t, T) = S_t e^{\int_t^T (f(t,u) - \delta(t,u))du} & G(t, T) = S_t e^{\int_t^T (f(t,u) - \epsilon(t,u))du} \end{array}$$

Why do Futures and Forwards have different convenience yields? →
Because $G < F$. G incorporates risk of having to cover daily margin calls.

Q-dynamics of this Miltersen and Schwarz model:

$$\begin{aligned} \frac{dS_t}{S_t} &= \mu_S(t)dt + \sigma_S(t) \cdot dW_t^Q, \\ df(t, T) &= \mu_f(t, T)dt + \sigma_f(t, T) \cdot dW_t^Q, \\ d\epsilon(t, T) &= \mu_\epsilon(t, T)dt + \sigma_\epsilon(t, T) \cdot dW_t^Q. \end{aligned}$$

$$\begin{aligned} \frac{dS_t}{S_t} &= [f(t, t) - \epsilon(t, t)]dt + \sigma_S(t) \cdot dW_t^Q, \\ df(t, T) &= \sigma_f(t, T) \cdot \left[\int_t^T \sigma_f(t, u)du \right] dt + \sigma_f(t, T) \cdot dW_t^Q, \\ d\epsilon(t, T) &= \left\{ \sigma_f(t, T) \cdot \left[\int_t^T \sigma_f(t, s)ds \right] \right. \\ &\quad \left. + [\sigma_f(t, T) - \sigma_\epsilon(t, T)] \cdot \left[\sigma_S(t) + \int_t^T (\sigma_f(t, s) - \sigma_\epsilon(t, s))ds \right] \right\} dt \\ &\quad + \sigma_\epsilon(t, T) \cdot dW_t^Q, \\ \frac{dG(t, T)}{G(t, T)} &= \left(\sigma_S(t) + \int_t^T (\sigma_f(t, s) - \sigma_\epsilon(t, s))ds \right) \cdot dW_t^Q = \sigma_G(t, T) \cdot dW_t^Q \end{aligned}$$

If we assume deterministic volatilities, there exists a closed-form solution using Black-76 formula.